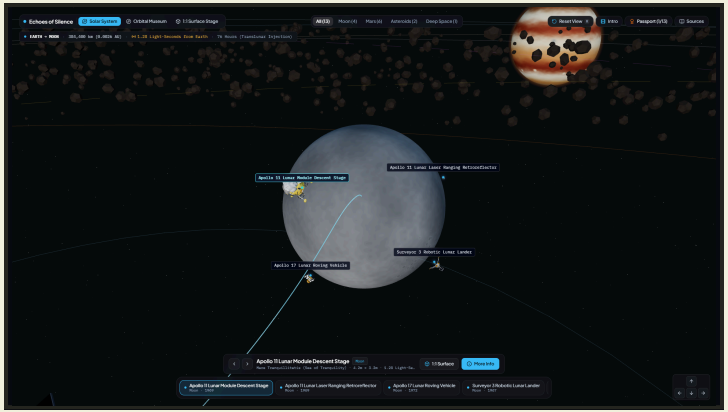
NASA SPACE APPS CHALLENGE CAIRO 2026

ECHOES OF SILENCE



Turning Silent NASA Spacecraft Across the Moon, Mars, and Deep Space into a Living 1:1 3D Museum

TEAM: SPAGHETTI CODERS

MEMBERS: Omar, Bashar, Youssef

github.com/omarsh4114/nasa-abandoned-not-forgotten-404

The Problem: Priceless Scientific Monuments

PIONEERS

Silent Pioneers Across 5 Worlds

Decades of exploration left landers and rovers on the Moon, Mars, Eros, Ceres, and Deep Space.

ENGAGEMENT

The Youth Disconnect (Ages 10–14)

Technical archives make historic machines feel like "space trash" to students.

SCALE

Invisible Scale & Distance

Students cannot picture true metric sizes or the vast light-delay across the system.

How do we turn silent NASA hardware into living, first-person teachers for the next generation of explorers?

"SPACE EXPLORATION IS A JOURNEY OF DISCOVERY, NOT A DESTINATION OF DEBRIS."

Our Solution: 3-Tier Interactive 3D Web Exhibition

MODE 1

Solar System Explorer

Trace mission trajectories (Translunar, Hohmann) and visualize real-time radio light-delay from Earth.

MODE 2

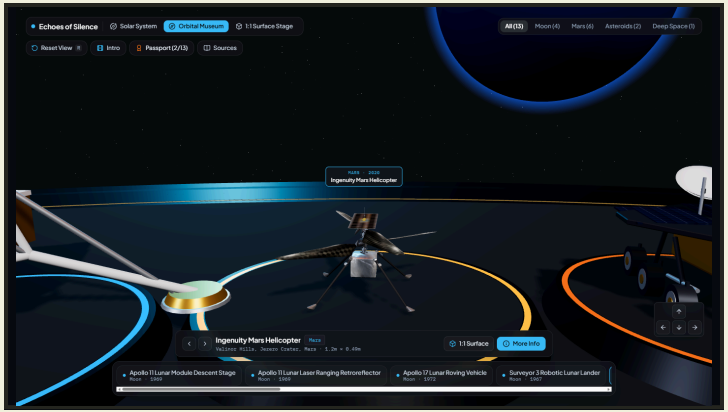
Orbital Museum Deck

Compare all 13 spacecraft side-by-side on an architectural exhibition platform.

MODE 3

1:1 Surface Stage & AR

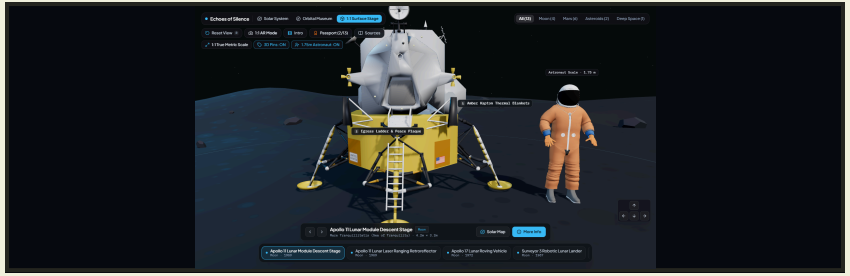
Walk around spacecraft at true metric scale next to an astronaut reference or in mobile AR.

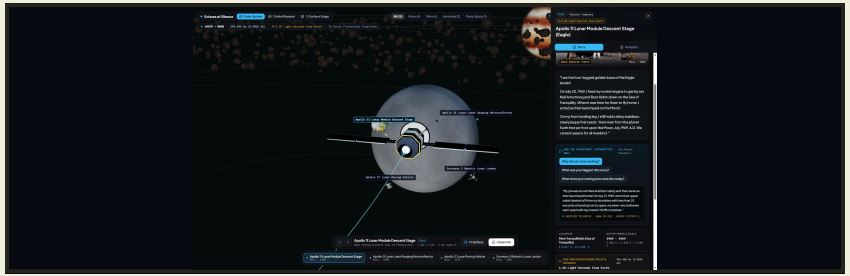


ARCHITECTURAL COMPARISON MODE

"CONNECTING THE PAST TO THE FUTURE THROUGH INTERACTIVE SPATIAL DATA."

How It Works: From 1:1 3D Inspection to Storytelling





STEP 01

Inspect at 1:1 Human Scale

Select spacecraft, toggle Astronaut reference, and click 3D engineering pins.

STEP 02

Compare "Then vs. Now"

Switch between historic mission photos and modern orbital recon (LROC/HiRISE).

STEP 03

"Ask the Spacecraft" Q&A

Click question pills to read first-person answers with verified NASA telemetry tags.

"TRANSFORMING ABSTRACT COORDINATES INTO FIRST-PERSON HUMAN STORYTELLING."

Powered by 6 Official NASA Open Data Repositories

NASA NSSDCA

Verified spacecraft catalog IDs, launch/end dates, and true metric dimensions for all 13 missions.

NASA 3D RESOURCES

Official glTF/GLB spacecraft geometry for Apollo LM, Ingenuity, InSight, Viking, and Dawn.

NASA LROC

Lunar coordinates and NAC orbital frames for Apollo 11 Eagle, LRV-3, and Surveyor 3.

MARS HIRISE

Martian coordinates and orbital imagery for Ingenuity, Opportunity, Spirit, and Sojourner.

NASA PDS

Geotechnical reports, InSight SEIS Marsquake logs (1,319 events), and Small Bodies Node data.

NASA IMAGE LIB

Public-domain mission photography and Kepler transit discovery records for 2,662 exoplanets.

Prototype Status & Tech Transparency

WHAT WORKS NOW

TECH STACK:

React 19, TypeScript, Three.js, React Three Fiber, GLSL Shaders, Tailwind CSS v4.

FUNCTIONAL:

- All 3 spatial modes & 13 unified exhibits
- 39 3D engineering pins & interactive cards
- One-way radio light-delay telemetry
- 1:1 WebRTC AR camera mode

LIMITS & DISCLOSURE

CURRENT LIMITS:

Artistic orbital anchors used for framing; 7 exhibits use procedural assemblies for fast loading.

AI DISCLOSURE:

Google Antigravity & Gemini assisted with 3D code/shaders; Manus AI structured this deck. Science data/photos are unmodified NASA records.

"WE CONCEIVED THE PROJECT INDEPENDENTLY; AI SERVED AS OUR TECHNICAL CO-PILOT."

Impact & Future Roadmap: Inspiring Space Archaeologists

EDUCATIONAL BENEFITS

For Students (Ages 10–14):

Transforms abstract telemetry into an emotional journey with rewards like the "Explorer Passport Certificate."

For Teachers & Museums:

Zero-install browser exhibition for any device with no login or data collection walls.

FUTURE ROADMAP

1. Live NASA JPL Horizons Ephemeris API: Real-time orbital dates.
2. Bilingual Arabic & English Mode: Full localization and worksheets.
3. Outer Solar System Expansion: Cassini, Huygens, Galileo, and Voyager.

"ECHOES OF SILENCE: ABANDONED BUT NOT FORGOTTEN."